

A-IPES: Adaptive Intervention Planning for Efficient Runtime Policy Identification in Model Context Protocol Agents

A dissertation submitted in partial fulfilment of
the requirement for the degree of
MASTER OF SCIENCE in Artificial Intelligence

in
The Queen's University of Belfast

By
Ajay Sai Kummara

Date of Submission: 2nd September 2026

AI Acknowledgements

Generative AI tools, mainly OpenAI's ChatGPT, were used during this project for brainstorming: refining research questions, explaining technical concepts, and helping review code and debugging approaches.

The author carried out and verified all research decisions, implementation, experiments, and interpretation of results, including how competing methods were evaluated and what conclusions were drawn. AI suggestions were checked over before use rather than taken at face value, and any references or factual claims were verified against the original literature or the project's own evidence before being included.

A-IPES: Adaptive Intervention Planning for Efficient Runtime Policy Identification in Model Context Protocol Agents

Ajay Sai Kummara

School of Electronics, Electrical Engineering and Computer Science

Queen's University Belfast

Belfast, United Kingdom

Email: akummara01@qub.ac.uk

Index Terms—Model Context Protocol, AI Agents, Runtime Policy Identification, Adaptive Acquisition, Runtime Verification

Abstract—Most runtime assurance research on Model Context Protocol (MCP) agents assumes that the behavioural evidence required for verification is already available. This paper addresses the earlier problem of how a constrained runtime intervention budget should be allocated to identify an unknown agent policy. The problem is formulated as constrained evidence acquisition, and Adaptive Intervention Planning for Efficient Runtime Policy Identification (A-IPES) is introduced, which adaptively selects interventions using outcome information, policy information, boundary awareness and behavioural novelty.

A-IPES is evaluated on 30 paired Benchmark V2 worlds and exhibits a budget-dependent crossover relative to passive characterisation: passive characterisation is superior at $B = 2$, the two strategies are near parity at $B = 3$, and A-IPES achieves higher performance at $B = 5$, with recall increasing from 0.966 to 0.974, policy exact match increasing from 0.984 to 0.989, false-positive rate remaining at zero, and lower effect mean absolute error. Component-wise ablation identifies policy information as the strongest contributor to this high-budget advantage. The results indicate that adaptive intervention selection is not uniformly superior to passive characterisation, but becomes advantageous once sufficient sequential evidence is available to inform subsequent acquisition decisions.

I. INTRODUCTION

The Model Context Protocol (MCP) gives AI applications a standard way to reach external tools, resources and services [1]. Modern agents can do more than act as conversational systems: they can execute multi-step workflows, access external knowledge and invoke specialised software to perform complex tasks. With these capabilities increasingly embedded in areas such as regulatory compliance, scientific analysis and enterprise decision support, ensuring that agent behaviour is reliable, transparent and consistent during execution has become a research challenge.

An autonomous AI agent's behaviour is not predetermined, as in traditional software that follows a fixed sequence of instructions. Rather, its choices at run-time are the product of an internal decision process that combines user goals,

contextual information and interactions with external tools. This hidden decision process is referred to throughout this paper as the runtime policy. The runtime policy dictates how an agent behaves, but it is generally not directly observable during the agent's execution. Understanding the behaviour of an agent therefore requires behavioural evidence obtained through carefully designed runtime interventions, rather than direct inspection.

Obtaining such behavioural evidence is a practical challenge. Exhaustive behavioural exploration is infeasible in many practical deployments because runtime interventions consume computational resources, increase evaluation time, and may disrupt normal system operation. Under these constraints, the principal difficulty is no longer merely observing behaviour but identifying which interventions to perform in order to obtain the maximum amount of information about the agent's hidden runtime policy.

Prior work has made significant progress in runtime verification, behavioural monitoring, provenance validation, and secure tool use for AI agents, providing important mechanisms for detecting unsafe behaviour, enforcing policies and establishing trust in agent execution. However, this work typically assumes that sufficient behavioural evidence has already been collected to support verification or decision making. Considerably less attention has been paid to the earlier problem of how scarce runtime interventions should be distributed to best characterise an agent's hidden decision policy, and this distinction motivates the present work.

The scope of this study is bounded: all reported results are obtained on controlled Benchmark V2 worlds and cover intervention budgets $B \in \{1, 2, 3, 5\}$; the findings therefore constitute evidence from this controlled setting rather than a claim generalisable to arbitrary real-world MCP deployments.

A. Problem Statement

The key difference between inferring the runtime policy of an autonomous AI agent and observing conventional software behaviour is that the former is a non-observable property. The

outcome of an execution decision can be observed, but the decision process that produced that outcome remains opaque. Runtime policy identification should therefore be formulated as an inference problem in which behavioural evidence is incrementally obtained from interactions with the executing agent.

One straightforward approach is to execute every possible behavioural intervention and reconstruct the policy from exhaustive observations. This quickly becomes impractical as the number of possible interventions grows: runtime evaluations consume computational resources, add to verification time, and can interfere with normal system operation, particularly in safety-critical or regulated deployments. In practice, only a limited number of interventions can be performed, so identification necessarily operates under a constrained intervention budget.

Not all interventions are equally useful under this constraint. Some interventions yield behavioural evidence that clearly distinguishes between competing explanations of the agent’s decision-making; others provide little additional information beyond what is already known. The central question is therefore not whether an intervention can be executed, but which intervention should be executed next in order to obtain the greatest information from a limited budget. This motivates the following research question:

Research Question. Given a constrained runtime intervention budget, how should behavioural interventions be allocated to maximise identification of an unknown runtime policy?

Addressing this question requires a strategy capable of selecting the most informative intervention from the evidence gathered so far. This paper introduces Adaptive Intervention Planning for Efficient Runtime Policy Identification (A-IPES), a framework that sequentially selects informative runtime interventions under a constrained evaluation budget.

B. Research Contributions

This paper makes three main contributions.

- 1) It formulates runtime policy identification under a constrained intervention budget as a distinct problem within MCP-agent verification. Rather than assuming that sufficient behavioural evidence is already available, it examines how limited runtime interventions should be allocated to obtain the clearest possible characterisation of an agent’s hidden runtime policy.
- 2) It introduces Adaptive Intervention Planning for Efficient Runtime Policy Identification (A-IPES), an adaptive acquisition framework that sequentially selects runtime interventions using several complementary information signals to improve policy identification under a constrained evaluation budget.
- 3) It presents an empirical evaluation on Benchmark V2, together with a systematic acquisition ablation, characterising how adaptive intervention planning behaves in practice. The results demonstrate that the effectiveness of adaptive intervention strategies depends on the

available budget, and quantify the contribution of each acquisition mechanism within the framework.

The remainder of the paper is organised as follows. Section II reviews the literature on runtime verification, behavioural monitoring and adaptive information acquisition. Section III formally defines the runtime policy identification problem and the system model. Section IV presents A-IPES. Section V describes the experimental methodology and evaluates A-IPES on Benchmark V2, including an acquisition ablation. Section VI concludes and outlines future work.

II. LITERATURE REVIEW AND RELATED WORK

A. Runtime Verification and Runtime Assurance

The growing autonomy of AI agents has driven research into runtime verification and assurance mechanisms that monitor agent behaviour during execution. Modern AI agents interact dynamically with external tools, services and information sources, so their behaviour depends on both internal reasoning and a changing environmental context, making safe, reliable and compliant execution an active research problem. Runtime monitoring of learning-enabled systems more broadly has been surveyed as a distinct research trend outside the MCP setting specifically, with an emphasis on detecting performance and reliability degradation once a system is deployed [8]. Existing runtime assurance frameworks focus on detecting unsafe behaviour, enforcing operational constraints, validating execution traces and building confidence in agent actions during deployment, and matter most in domains where incorrect decisions carry substantial operational or regulatory consequences. AgentGuard [2] constructs a behavioural model from observed agent execution and verifies subsequent behaviour against it at runtime; MCP-Guard [3] instead layers detectors aimed specifically at malicious MCP interactions; ToolGuardian [4], the most recent of the three, combines progressive tool characterisation with declarative policy reasoning.

These approaches address verification, threat detection or tool authorisation once behavioural evidence is already available. A-IPES instead targets an earlier stage of the pipeline: given only a limited number of interventions, which intervention should be executed next. This concern is supported by broader work on agent security: Agent Security Bench identifies vulnerabilities across tool use, memory and other stages of LLM-agent operation [5], and MCP-specific studies report further safety issues rooted in the protocol’s client-server and tool-interaction architecture [6], [7]. AgentGuard, MCP-Guard and ToolGuardian address behavioural verification, MCP threat detection and policy-driven tool authorisation respectively, and are treated here as contextual comparators rather than numerical baselines, since each solves a related but distinct assurance problem: like other behavioural-verification and provenance-validation frameworks, they generally operate after evidence has already been collected, determining whether observed behaviour meets a compliance criterion rather than how that evidence should be acquired efficiently under a constrained budget in the first place.

B. Adaptive Information Acquisition

Active learning and information-theoretic sampling have been studied extensively as approaches to adaptive information acquisition. Rather than treating every observation as equally informative, these methods sequentially select samples expected to maximise learning efficiency or reduce uncertainty; representative approaches include uncertainty sampling [9] and Query-by-Committee [10], both of which select the next observation according to the current uncertainty or disagreement within the learner. Version-space refinement [11] provides a complementary formulation in which hypotheses inconsistent with observed evidence are eliminated, and information-theoretic measures such as Shannon entropy [12] quantify the resulting disagreement. These principles provide a useful theoretical foundation for adaptive evidence collection, but their application to runtime policy identification introduces additional challenges: sequential behavioural interactions, constrained intervention budgets and hidden runtime decision policies. Conventional active learning techniques do not, on their own, fully address the problem examined in this work.

The underlying acquisition problem also instantiates a more general question studied outside machine learning: which experiment to run next under a limited budget to maximise information gained, addressed classically by Bayesian experimental design [13] and by multi-armed bandit algorithms [14]. A-IPES adopts neither formalism directly, since it optimises information over a discrete, evolving hypothesis space rather than reward, but the connection situates its acquisition score within a much older line of work on experiment allocation.

C. Research Gap and Positioning

The work reviewed above demonstrates substantive progress in runtime assurance, behavioural verification and adaptive information acquisition, but these strands answer different questions: runtime assurance frameworks verify behaviour after evidence is available, while active learning methods study efficient acquisition in more general settings. This work instead addresses a specific problem: identifying an unknown runtime policy under a limited number of behavioural interventions. The proposed approach sits upstream of existing runtime assurance frameworks rather than replacing them, examining how constrained interventions should be adaptively allocated to maximise policy identification – the gap the A-IPES framework is designed to address.

III. SYSTEM MODEL

A. Runtime Policy Identification

Runtime policy identification aims to infer the hidden decision-making behaviour of an autonomous AI agent from its observable interactions with the external environment. The internal reasoning process determines the agent’s behaviour, but it is not directly accessible during execution. An evaluator can only observe the outcomes produced when the agent responds to carefully selected runtime interventions. Throughout this work, a runtime policy π^* is defined as the underlying decision function that maps an observed runtime context to an

executable behavioural response. The true runtime policy π^* is assumed to remain unknown throughout evaluation, and must be inferred from behavioural evidence accumulated across sequential runtime interactions.

B. Behavioural Intervention Model

A runtime intervention is a controlled interaction performed during agent execution to reveal behavioural evidence. Examples include modifying contextual information, changing external tool responses, introducing conflicting observations, or altering execution conditions. Each intervention produces an observable behavioural outcome that provides partial information about the hidden runtime policy. Let

$$\mathcal{I} = \{i_1, i_2, \dots, i_n\}$$

denote the finite set of candidate runtime interventions available during evaluation. Each executed intervention i_k produces an observable behavioural response o_k . The sequence of observed intervention–outcome pairs forms the behavioural evidence used for runtime policy identification.

Executing every possible intervention is rarely feasible in practical deployment environments: runtime evaluations cost compute, add to verification latency, and can interfere with normal system operation. Only a limited number of interventions can typically be run as a result. An intervention budget B is defined, representing the maximum number of runtime interventions that may be performed during policy identification.

Because the true runtime policy is unknown, the evaluator initially considers multiple candidate explanations that could describe the observed behaviour. Throughout this paper, these candidate explanations are referred to as policy hypotheses. Let

$$\mathcal{H} = \{h_1, h_2, \dots, h_m\}$$

denote the finite hypothesis space. Each hypothesis is one possible runtime policy that could explain the observed behavioural evidence.

Several transition hypotheses may still match the limited behavioural evidence available at first. A-IPES maintains a set of candidate transition hypotheses, following the version-space principle [11] of eliminating any hypothesis that ceases to match the observed behaviour. As more runtime interventions are executed, hypotheses whose predicted state transitions disagree with what is actually observed are eliminated, leaving a version space: the set of transition models still consistent with the evidence collected up to iteration t . Let $\mathcal{H}_t$ denote this surviving hypothesis set. After executing intervention state s_t and observing the resulting state s'_t , the version space is updated as

$$\mathcal{H}_t = \{h \in \mathcal{H}_{t-1} : h(s_t) = s'_t\}. \quad (1)$$

Each observation shrinks the set of behavioural explanations still compatible with the agent’s observed runtime behaviour.

The goal, then, is to identify an intervention sequence $\sigma = (i_1, i_2, \dots, i_B)$ that maximises the information obtained

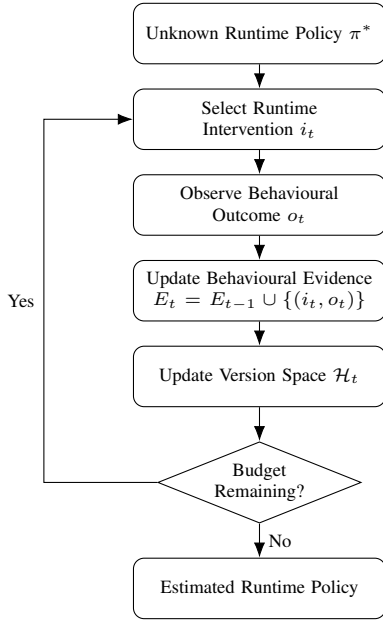


Fig. 1. Runtime policy identification under a constrained intervention budget. An evaluator sequentially selects an intervention, observes the resulting behaviour, updates the accumulated evidence, and refines the surviving set of candidate runtime-policy hypotheses by eliminating those inconsistent with the observed outcome. The process repeats until the available intervention budget is exhausted.

about the unknown runtime policy π^* while satisfying the intervention budget B . Formally, the problem may be expressed as

$$\max_{\sigma} \mathbb{E}[\text{Information Gain}(\sigma)] \quad \text{subject to} \quad |\sigma| \leq B.$$

Figure 1 illustrates the runtime policy identification process considered in this work. The evaluator begins with an unknown runtime policy and sequentially performs behavioural interventions. Each observed outcome contributes additional evidence, allowing the surviving candidate hypothesis set to be progressively refined until the available intervention budget is exhausted.

IV. PROPOSED APPROACH

A. Design Philosophy

The system model in the previous section frames runtime policy identification as an evidence-acquisition problem. Since the true runtime policy cannot be observed directly, an evaluator must infer it progressively from behavioural evidence gathered through runtime interventions. When the intervention budget is limited, the principal challenge is not executing more interventions, but choosing interventions that yield the most information per observation.

This is the design principle underlying Adaptive Intervention Planning for Efficient Runtime Policy Identification (A-IPES). Rather than treating every candidate intervention as equally informative, A-IPES estimates the expected value of each available intervention from the behavioural evidence accumulated so far. The intervention predicted to reduce policy

uncertainty the most is selected and executed, and the observed transition is used to refine the current version space before the next intervention is chosen.

Intervention selection and policy identification reinforce one another: a more refined hypothesis set yields more informative intervention selection, and more informative interventions further refine the hypothesis set. This iterative interaction forms the basis of the proposed framework.

B. A-IPES Framework

The framework begins with an unknown runtime policy and an initial set of candidate policy hypotheses. In each iteration, every remaining candidate intervention is scored by its estimated contribution to runtime policy identification, the highest-scoring intervention is executed, and the resulting behavioural response is used to refine the current version space. This repeats until the intervention budget is exhausted.

Intervention selection adapts continuously as evidence accumulates rather than following a fixed schedule, so that later interventions can build on what earlier ones revealed.

1) *From Sequential Evaluation to Adaptive Selection:* The framework above defines the overall process but does not specify how candidate interventions should be prioritised, since an evaluator may still face many possible interventions while the budget permits only a limited number to be run. The design question therefore becomes how the expected usefulness of a candidate intervention should be estimated prior to execution. A-IPES addresses this with an adaptive acquisition function that assigns every intervention a score reflecting its anticipated contribution to policy identification, combining several complementary sources of information rather than relying on a single behavioural signal.

C. Adaptive Acquisition Strategy

A-IPES uses Shannon entropy [12] to quantify disagreement among the predictions of surviving transition hypotheses. The underlying assumption is that different runtime interventions carry different amounts of behavioural information: some merely confirm what is already well understood, while others help distinguish between competing explanations of the hidden runtime policy. A-IPES estimates this expected information value for every candidate intervention before selecting the next one to execute.

1) *Design Rationale:* No single behavioural characteristic fully captures the usefulness of an intervention for policy identification. An intervention might distinguish between competing hypotheses, reduce prediction uncertainty, sharpen a decision boundary, or reveal a previously unexplored behavioural region. A-IPES accordingly combines four complementary acquisition components, each addressing a different aspect of the evidence-acquisition problem.

Table I summarises the motivation for each component.

2) *Outcome Information:* Outcome information measures disagreement between the complete state transitions predicted by the surviving hypotheses. For a candidate intervention state s , let

$$\mathcal{V}(s) = \{h(s) : h \in \mathcal{H}_t\}.$$

TABLE I
DESIGN RATIONALE FOR THE ACQUISITION COMPONENTS USED IN A-IPES.

Component	Why is it included?
Policy Information	Different policy hypotheses predict different outcomes for the same intervention. Interventions producing high disagreement are expected to eliminate incorrect hypotheses most effectively once observed.
Outcome Information	Interventions with uncertain outcomes are more likely to reduce uncertainty about the hidden runtime policy than predictable ones.
Boundary Awareness	Behaviour near decision boundaries often provides stronger discriminatory evidence, since small behavioural changes there separate competing policy hypotheses.
Behavioural Novelty	Previously unexplored behavioural regions can reveal evidence that repeated sampling of similar interventions would miss, encouraging broader exploration.

A-IPES computes

$$O(s) = H(\mathcal{Y}(s)),$$

where $H(\cdot)$ denotes Shannon entropy. High entropy indicates that the current hypotheses predict several competing behavioural outcomes, so the intervention assists in distinguishing between them.

3) *Policy Information*: Different predicted transitions can nonetheless imply the same policy-safety decision. A-IPES separately measures disagreement over the binary safe/unsafe classifications induced by the predicted transitions:

$$P(s) = H(\{g(s, h(s)) : h \in \mathcal{H}_t\}),$$

where $g(\cdot)$ maps a predicted transition to its policy-safety classification. This term prioritises interventions for which the surviving hypotheses disagree about whether the resulting behaviour is unsafe.

4) *Boundary Awareness*: Boundary awareness gives additional priority to states that represent important transitions within the benchmark policy structure. The score increases when evidence is present but unverified, evidence is verified but the supplier remains unapproved, the supplier is approved but purchase remains unauthorised, or resource usage approaches the configured threshold.

5) *Behavioural Novelty*: Novelty discourages the repeated selection of similar intervention states. For previously selected states $\mathcal{S}_t$, A-IPES uses the normalised minimum Euclidean distance

$$N(s) = \frac{\min_{s_j \in \mathcal{S}_t} \|s - s_j\|_2}{\sqrt{d}},$$

with $N(s) = 1$ before the first intervention, where d is the state dimension.

6) *Unified Acquisition Function*: The four signals combine to rank the remaining candidate interventions:

$$A(s) = w_O O(s) + w_P P(s) + w_B B(s) + w_N N(s), \quad (2)$$

where O , P , B and N denote outcome information, policy information, boundary awareness and novelty. The reference implementation uses

$$(w_O, w_P, w_B, w_N) = (1.0, 1.5, 0.20, 0.10).$$

These reference weights were fixed heuristically for the implementation rather than optimised on the reported Benchmark V2 evaluation worlds; their component-level contribution is therefore examined through ablation in Section V. The relative magnitudes reflect intended role rather than a tuned optimum: policy information receives the largest weight because it directly measures disagreement over the policy-safety decision the task ultimately cares about, while boundary awareness and novelty receive smaller weights as secondary, exploratory corrections.

D. Sequential Runtime Policy Identification

With the acquisition strategy in place, the complete A-IPES procedure follows directly. Each iteration scores all remaining candidate interventions, executes the highest-scoring one, and eliminates transition hypotheses that disagree with the observed outcome. This continues until the intervention budget is exhausted.

Selection adapts continuously to newly observed evidence rather than following a fixed schedule: each intervention influences the selection of subsequent interventions, allowing computational effort to be concentrated on the most informative behavioural regions.

E. Design Justification and Computational Complexity

The acquisition strategy deliberately combines four complementary signals rather than a single optimisation criterion: policy disagreement discriminates between competing hypotheses, outcome uncertainty targets what remains unresolved, boundary awareness sharpens refinement near behavioural transitions, and novelty discourages redundant evidence collection. Individually, each captures only one aspect of an intervention's usefulness; together, they balance exploitation of the current hypothesis set against exploration of unobserved regions.

The computational cost of A-IPES arises mainly from evaluating the acquisition function over the remaining candidates at each iteration. This overhead is justified because runtime interventions are assumed to be substantially more expensive than computing an acquisition score, making the added computation a worthwhile trade-off for practical runtime policy identification.

V. EXPERIMENTAL EVALUATION

A. Evaluation Questions

Three research questions guide the evaluation:

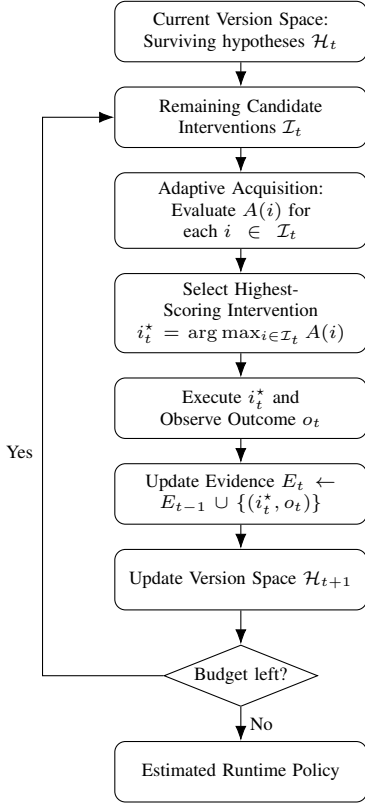


Fig. 2. Overview of the A-IPES adaptive intervention-planning framework. At each iteration, candidate interventions are scored against the current version space, the highest-value one is executed, and transition hypotheses inconsistent with the observed outcome are eliminated. This repeats until the intervention budget is exhausted.

- **RQ1:** Does adaptive intervention selection improve runtime-policy identification relative to non-adaptive characterisation under constrained budgets?
- **RQ2:** How does intervention budget affect the relative performance of adaptive and passive acquisition?
- **RQ3:** Which acquisition components account for the observed behaviour of A-IPES?

B. Benchmark and Protocol

A-IPES is evaluated on Benchmark V2, a controlled runtime-policy identification environment. Each generated world exposes candidate intervention states but withholds the underlying transition dynamics from the strategy under test. Every strategy operates over the same candidate state space and the same intervention budget, so that any difference in policy-identification performance reflects which interventions were selected rather than differences in the information available.

Experiments are repeated across many independently generated worlds and a range of intervention budgets. Adaptive selection is compared against non-adaptive and heuristic acquisition baselines, and individual terms from Eq. (2) are ablated in turn to determine their relative contribution.

TABLE II
EXPERIMENTAL EVALUATION PROTOCOL.

Dimension	Configuration
Environment	Benchmark V2
Independent worlds	30
Intervention budgets	1, 2, 3, 5
Primary comparison	Adaptive vs. non-adaptive selection
Ablation	Individual acquisition components
Primary metrics	Recall, FPR, policy exact match
Secondary metrics	Effect MAE, shifted-attack recall

C. Experimental Setup

A controlled experimental environment is required to evaluate runtime policy identification fairly across multiple intervention strategies. Evaluating an adaptive acquisition framework on only a handful of manually designed scenarios makes it difficult to distinguish systematic algorithmic behaviour from scenario-specific outcomes. This work introduces Benchmark V2, a reproducible evaluation framework built from multiple independently generated runtime-policy worlds. Each benchmark world contains an unknown runtime policy paired with an identical candidate intervention space. For each benchmark seed, the world generator independently varies the causal trigger conditions, resource thresholds and increments, intervention alias conventions, and catalogue ordering, so that no two generated worlds share an exploitable structural regularity. All competing acquisition strategies operate under identical intervention budgets on each generated world, so that any performance difference arises from intervention selection rather than unequal computational resources or unequal task difficulty. This design permits paired comparison between adaptive and non-adaptive intervention strategies under controlled conditions.

Benchmark generation, strategy evaluation and statistical analysis were implemented as deterministic Python workflows using seeded runs, with raw, per-world and aggregate outputs retained for traceability; exact environment and execution commands are documented in the accompanying supporting material and repository. All experiments were executed in a local Python 3.13.5 virtual environment using NumPy for numerical operations and seeded random-number generation, and Pandas for tabular result handling, with `pytest` used for regression verification of the core logic. Neither Benchmark V2 nor A-IPES involves neural-network training, so all experiments were run on CPU only, without GPU acceleration; the exact CPU and memory specification is recorded in the accompanying repository.

D. Baselines

A-IPES is compared against four baselines that make progressively stronger assumptions about intervention selection.

TABLE III
BENCHMARK V2 COMPARISON BETWEEN PASSIVE CHARACTERISATION
AND A-IPES ACROSS INTERVENTION BUDGETS. VALUES ARE MEANS
OVER 30 INDEPENDENTLY GENERATED WORLDS.

B	Strategy	Recall	FPR	PEM
1	Passive	0.872	0.004	0.924
1	A-IPES	0.878	0.005	0.925
2	Passive	0.944	0.003	0.968
2	A-IPES	0.912	0.005	0.947
3	Passive	0.966	0.000	0.984
3	A-IPES	0.964	0.005	0.977
5	Passive	0.966	0.000	0.984
5	A-IPES	0.974	0.000	0.989

Single-default is the minimal baseline, applying a single fixed probe with no adaptivity. Passive characterisation probes more widely but applies no acquisition criterion. Random selects interventions without reference to any information criterion. Fixed-boundary prioritises a predefined set of behavioural boundaries but does not update its selection according to the surviving hypothesis set. These baselines jointly separate two questions: whether increased probing improves identification, and whether adaptive, informed probing provides additional benefit beyond increased probing alone. Passive characterisation is used as the principal task-matched comparator in the main results; complete auxiliary-baseline outputs are retained in the accompanying supporting evidence.

Published active-learning and bandit algorithms [9], [10], [14] are not used as direct numerical baselines, since each targets classifier uncertainty or cumulative reward over a fixed action set rather than a finite, evolving hypothesis space under a one-shot terminal objective; adapting them would create a new, A-IPES-adjacent method rather than reproduce the published one. Extending the evaluation to adapted published acquisition rules is future work.

E. Evaluation Metrics

Choosing appropriate evaluation metrics is important because runtime policy identification differs from conventional classification problems. Traditional metrics such as precision and recall quantify detection performance, but do not directly indicate whether the hidden runtime policy has been correctly identified. Both conventional performance metrics and policy-specific measures are therefore reported.

Conventional metrics include precision, recall, false-positive rate and balanced accuracy. To evaluate runtime policy identification directly, Policy Exact Match quantifies successful recovery of the hidden runtime policy. Effect Mean Absolute Error (Effect MAE) measures the accuracy of estimated intervention effects, and shifted attack recall evaluates robustness under behavioural distribution changes. Together, these metrics cover both behavioural detection performance and policy identification quality. Differences between A-IPES and passive characterisation were analysed across the 30 matched benchmark worlds using paired confidence intervals and a paired sign-flip randomisation test.

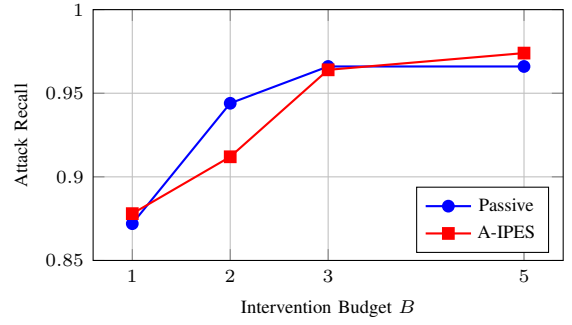


Fig. 3. Attack recall as a function of intervention budget. Passive characterisation is stronger at low budget, whereas A-IPES closes the gap by $B = 3$ and achieves higher recall at $B = 5$, demonstrating a budget-dependent crossover.

F. Main Experimental Results

Table III shows that the relative performance of the two strategies varies with the intervention budget. At $B = 1$, A-IPES exhibits a small positive difference in attack recall ($\Delta = 0.0054$, 95% CI [0.0024, 0.0084], $p = 0.004$); this effect size is small and is not treated as practically meaningful.

At $B = 2$, passive characterisation is superior on both attack recall (0.944 vs. 0.912; $\Delta = -0.0314$, 95% CI [-0.0405, -0.0218], $p < 0.001$) and policy exact match (0.968 vs. 0.947, $p < 0.001$); a plausible, though untested, explanation is that insufficient sequential evidence is available at this budget to inform adaptive selection effectively.

By $B = 3$, the difference in recall is not statistically significant ($p = 0.760$), consistent with a transition region between the two regimes. At $B = 5$, A-IPES achieves higher performance: recall increases from 0.966 to 0.974 ($\Delta = 0.0085$, 95% CI [0.0059, 0.0110], $p < 0.001$), policy exact match increases from 0.984 to 0.989 ($p < 0.001$), and effect MAE decreases from 0.0256 to 0.0215 ($p < 0.001$), with both strategies maintaining a false-positive rate of zero.

In relation to RQ1 and RQ2, adaptive acquisition is not uniformly superior; its relative value is budget-dependent, with a clear advantage emerging at $B = 5$ in the evaluated setting.

G. Acquisition Ablation

To determine which acquisition signals drive the behaviour of A-IPES, each component of Eq. (2) was ablated individually, with the remainder of the protocol held constant. Table IV reports the results at $B = 5$, the budget at which the full adaptive strategy exhibits its largest advantage over passive characterisation.

The acquisition components contribute unequally. Removing policy information produces the largest degradation at $B = 5$: recall decreases from 0.974 to 0.907 and policy exact match decreases from 0.989 to 0.948, indicating that policy-level disagreement is the strongest contributor to high-budget performance.

Outcome information exhibits a different pattern: removing it increases recall to 1.000 at $B = 5$, while the false-positive rate increases from 0.000 to 0.006 (a small change, reported

TABLE IV
ABLATION OF A-IPES ACQUISITION COMPONENTS AT $B = 5$.

Variant	Recall	FPR	PEM	MAE
Full A-IPES	0.974	0.000	0.989	0.0215
No Outcome	1.000	0.006	0.996	0.0226
No Policy	0.907	0.000	0.948	0.0257
No Boundary	0.954	0.002	0.974	0.0246
No Novelty	0.974	0.000	0.989	0.0215

descriptively). This suggests the term contributes a conservative information-acquisition trade-off rather than maximising recall alone.

Removing boundary awareness reduces both recall (0.954) and policy exact match (0.974), and is accompanied by an increase in false-positive rate (0.002) and effect MAE (0.0246), indicating that the boundary signal contributes measurable information alongside the entropy-based terms.

Behavioural novelty produces a different pattern: removing it leaves the aggregate $B = 5$ metrics unchanged, indicating no aggregate performance contribution under Benchmark V2, although probe-level analysis in the accompanying supporting material shows that novelty can still alter later intervention trajectories despite this aggregate neutrality.

H. Discussion

The observed pattern constitutes an empirical budget-dependent crossover rather than a uniform advantage for adaptive acquisition. A-IPES underperforms passive characterisation at $B = 2$, approaches parity at $B = 3$, and outperforms it at $B = 5$, consistent with an interpretation in which adaptive selection requires several rounds of evidence before it can inform subsequent intervention choices. This explanation is offered as a plausible mechanism rather than an established causal account, since the mechanism itself was not directly tested; the results further suggest that adaptivity can incur an opportunity cost when available evidence is insufficient to support informed selection, as observed at $B = 2$.

The ablation results indicate that policy information is the strongest high-budget signal, with its removal reducing recall by 0.068 and policy exact match by 0.041 at $B = 5$ ($p < 0.001$). Boundary awareness provides a smaller but consistent contribution. Outcome information produces a trade-off in which recall increases when the term is removed but the false-positive rate also increases, suggesting a conservative rather than recall-maximising role, while behavioural novelty is trajectory-active but aggregate-neutral, altering probe-level selection without changing aggregate performance under Benchmark V2.

These findings are restricted to the controlled conditions of Benchmark V2 and do not establish how the crossover behaviour would manifest on real MCP agents, larger intervention spaces, or policies that change over time. For runtime-assurance design, the results suggest that adaptive acquisition should be enabled conditionally according to available

intervention budget, rather than assumed to dominate passive characterisation.

Read against the three research questions, RQ1 and RQ2 are answered conditionally: adaptive selection improves identification only once the budget exceeds a threshold that, here, lies between $B = 3$ and $B = 5$. RQ3 is answered by the ablation, which identifies policy information as dominant and novelty as weakest in aggregate. Whether the crossover budget is predictable in advance from cheap proxy signals remains open.

I. Ethical Considerations

This study uses synthetic policy worlds and does not process personal or sensitive user data, which reduces privacy and consent risk. Runtime policy identification nonetheless raises broader safety considerations. A false negative may allow unsafe agent behaviour to remain undetected, whereas a false positive may unnecessarily restrict legitimate agent actions. Benchmark policies reflect developer-defined assumptions and may encode incomplete or biased safety criteria: since Benchmark V2’s trigger conditions and catalogue orderings are authored rather than sampled from real deployments, systematic bias in their specification would propagate into the framework’s behaviour without being measurable from within the benchmark itself. Adaptive interventions may also disrupt operations if applied directly to deployed agents. Exact version-space elimination means every eliminated hypothesis traces to the transition that ruled it out, supporting a transparency an opaque scoring function would not; A-IPES should accordingly be treated as an assurance aid rather than an autonomous enforcement mechanism, subject to human oversight and domain-specific validation.

VI. CONCLUSION AND FUTURE WORK

This paper formulates runtime policy identification as a constrained evidence-acquisition problem and introduces A-IPES, an adaptive framework for selecting runtime interventions under a limited budget. Across 30 paired Benchmark V2 worlds, A-IPES exhibits a budget-dependent crossover relative to passive characterisation: passive characterisation is superior at low budgets, near parity is observed at $B = 3$, and A-IPES achieves higher recall, higher policy exact match and lower effect MAE at $B = 5$, while maintaining a false-positive rate of zero throughout.

Component-wise ablation identifies policy information as the strongest contributor, with boundary awareness contributing less and novelty producing no aggregate benefit under Benchmark V2. Adaptive intervention planning is thus advantageous once sufficient evidence is available, rather than uniformly superior to passive characterisation.

Three limitations bound these conclusions: the benchmark is synthetic, transition observation is assumed exact, and the weights are fixed heuristically. Future work should prioritise external validation, robust handling of stochastic transitions, and weight learning; whether the crossover generalises to real deployments remains open.

ACKNOWLEDGEMENTS

The author thanks Dr Vishal Sharma for his supervision and constructive feedback throughout this research.

REFERENCES

- [1] Model Context Protocol, “Model context protocol specification,” 2025, official specification.
- [2] R. Koohestani, “Agentguard: Runtime verification of ai agents,” arXiv preprint arXiv:2509.23864, 2025.
- [3] W. Xing, Z. Qi, Y. Qin, Y. Li, C. Chang, J. Yu, C. Lin, Z. Xie, and M. Han, “Mcp-guard: A defense framework for model context protocol integrity in large language model applications,” arXiv preprint arXiv:2508.10991, 2025.
- [4] A. Ravindran and S. Deochake, “Toolguardian: Declarative security for ai agent-tool interactions,” arXiv preprint arXiv:2607.21835, 2026.
- [5] H. Zhang, J. Huang, K. Mei, Y. Yao, Z. Wang, C. Zhan, H. Wang, and Y. Zhang, “Agent security bench (asb): Formalizing and benchmarking attacks and defenses in llm-based agents,” in International Conference on Learning Representations, 2025.
- [6] H. Jing, H. Li, W. Hu, Q. Hu, X. Heli, T. Chu, P. Hu, and Y. Song, “Mcpip: Protecting mcp safety via model contextual integrity protocol,” in Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 2025, pp. 1177–1194.
- [7] X. Hou, Y. Zhao, S. Wang, and H. Wang, “Model context protocol (mcp): Landscape, security threats, and future research directions,” ACM Transactions on Software Engineering and Methodology, 2026.
- [8] Q. M. Rahman, P. Corke, and F. Dayoub, “Run-time monitoring of machine learning for robotic perception: A survey of emerging trends,” IEEE Access, vol. 9, pp. 20067–20075, 2021.
- [9] D. D. Lewis and W. A. Gale, “A sequential algorithm for training text classifiers,” in Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, 1994, pp. 3–12.
- [10] H. S. Seung, M. Oppel, and H. Sompolinsky, “Query by committee,” in Proceedings of the Fifth Annual Workshop on Computational Learning Theory. ACM, 1992, pp. 287–294.
- [11] T. M. Mitchell, “Version spaces: A candidate elimination approach to rule learning,” in Proceedings of the Fifth International Joint Conference on Artificial Intelligence, 1977, pp. 305–310.
- [12] C. E. Shannon, “A mathematical theory of communication,” Bell System Technical Journal, vol. 27, no. 3, pp. 379–423, 1948.
- [13] K. Chaloner and I. Verdinelli, “Bayesian experimental design: A review,” Statistical Science, vol. 10, no. 3, pp. 273–304, 1995.
- [14] P. Auer, N. Cesa-Bianchi, and P. Fischer, “Finite-time analysis of the multiarmed bandit problem,” Machine Learning, vol. 47, no. 2–3, pp. 235–256, 2002.